Apr. 30

- first 30 minutes working on PSP
- one day extension on this week's HW to cover material here
- Course evaluations are open, sent to email

- When quotienting by an ideal, things in the ideal become zero

Ex. $\mathbb{R}[x]/\langle x^2+1\rangle \cong \mathbb{C}$

$x^2+1+\langle x^2+1\rangle = 0+\langle x^2+1\rangle \Rightarrow x^2+\langle x^2+1\rangle = -1+\langle x^2+1\rangle \Rightarrow x^2=-1$

x is behaving like i

$x^3-\pi x^2+x-77+\langle x^2+1\rangle \Rightarrow (-1)^2\cdot x-\pi\cdot(-1)+x-77+\langle x^2+1\rangle$

$= 2x+\pi-77+\langle x^2+1\rangle$

★ for HW q. about $\mathbb{Z}[x]/\langle 3, x^2+1\rangle$, think 3 is 0 and x^2+1 is $0 \to x^2=-1$

$7x^3-2x^2+5x+10+\langle 3, x^2+1\rangle$

$\Rightarrow 1\cdot x^3-2x^2+2x+1+\langle 3, x^2+1\rangle$

$\Rightarrow (1\cdot(-1)\cdot x)-2(-1)+2x+1+\langle 3, x^2+1\rangle$

$\Rightarrow -x+2+2x+1+\langle 3, x^2+1\rangle$

$\Rightarrow x+1+\langle 3, x^2+1\rangle$

Prime Ideals and Maximal Ideals

- Euclid's Lemma p is prime (positive integers) if and only if whenever p divides a product ab, $p|a$ or $p|b$.

inclusive or, could divide both

Useful notion from this: "to contain is to divide"

- Def. A proper ideal P in a commutative ring R is called a prime ideal if whenever $ab \in P$, then $a \in P$ or $b \in P$

$P \subsetneq R$

- Def. A maximal ideal M of a commutative ring R is a proper ideal of R such that whenever M is contained in an ideal I of R, then $I=M$ or $I=R$
  - $M \subsetneq I \subsetneq R$ is impossible
  - If M contains unity, $M=R$, otherwise $M=I$

- Thm. 14.3 Let R be a commutative ring with 1, and let A be an ideal of R. Then R/A is an integral domain $\Leftrightarrow$ A is prime

- Thm. 14.4 Let R be a commutative ring with 1, and let A be an ideal of R. Then R/A is a field $\Leftrightarrow$ A is maximal